

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	549052
Award Program	
Project Title	Evaluation of Best Management Practices for Bee & Monarch Habitat
Grant Start Date	01/01/2018
Grant End Date	12/31/2023
Total Grant Budget	\$717,612.00
Total FFAR Budget	\$353,044.00
Matching Funder(s)	Missouri Department of Conservation University of Missouri
Final Report Date	04/01/2024
Project Director/Principal Investigator Information	
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Columbia, MO, MO-003
- 1.2. How many new jobs were created because of this grant funding?

13 jobs were created (graduate student assistants, undergraduate technicians, and research technicians)
- 1.3. How many jobs were maintained because of this grant funding?

0 jobs were maintained
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No changes to tax status
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No changes to the name of the organization

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is



intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Establishing high quality non-crop habitat patches for insect pollinators can mitigate the adverse effects of landscape change due to urbanization and agricultural intensification. However, it is still unclear which aspects of non-crop habitat are the most important for increasing the ecological fitness of pollinator populations, and current recommendations by governmental and private organizations regarding seed mixture composition and management intensity are inconsistent. Given that native seed can be costly and sometimes difficult to obtain, it is critical that habitat planting recommendations are supported by rigorous scientific data. Working in collaboration with the Missouri Department of Conservation, our goal was to provide scientifically sound recommendations on native habitat plantings that maximize the benefits for pollinator populations while minimizing the associated economic costs. Our objectives were to: (1) improve the success of pollinator habitat plantings by developing recommendations on seed origin, seed mix diversity, planting density, and location of habitat to maximize the benefits for monarch butterflies and bumble bee populations, (2) determine proper on-going management of critical milkweed and nectar species for optimal plant establishment and growth to maximize survival and reproduction of monarch butterflies and bumble bees, (3) engage Master Gardeners and Master Naturalists across Missouri to observe and record utilization of native plant species by monarchs and bumble bees to further our understanding of the resource requirements for these species in Missouri, and (4) communicate knowledge gained to homeowners, landowners, and agricultural consultants to encourage community support for enhancing pollinator habitat. Our data suggests that pollinator habitats can be planted and managed in ways that enhance the utility of habitats for monarchs and bumble bees. Use of seed from adjacent regions and management through mowing may extend the time that resources are available to pollinators without disrupting synchrony. Milkweed planted near shady, wooded areas may benefit adult monarchs without increasing the risk of predation on monarch eggs. Milkweed plants within a matrix of diverse flowering nectar plants are equally apparent to monarch females as those in stand-alone plots. These findings will increase community engagement in pollinator habitat efforts by providing recommendations supported by scientific and observational data from across Missouri. As a result, this project will significantly enhance the ecological fitness of pollinator populations, particularly monarchs and bumble bees, due to the greater abundance of high-quality habitats available across Missouri.



- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Wild insect pollinators supply an estimated \$3 billion annually in pollination services in the US. However, native pollinators have declined markedly in recent decades with habitat loss and land use intensification highlighted as leading factors. Establishing high quality non-crop habitat patches for insect pollinators can mitigate the adverse effects of broad-scale landscape change. Our work documents that native wildflower and milkweed plantings can enhance pollinator diversity and abundance in a row crop-dominated landscape by providing food resources and nesting sites. We found that several habitat characteristics are particularly important for increasing the ecological fitness of pollinator populations, including seed source (regional genetic differences), management through mowing, and the configuration of the habitat within the broader landscape. The US Farm Bill has invested in a range of incentive-based conservation programs on agricultural land to mitigate pollinator declines while providing multiple other benefits for agriculture and surrounding environments. Our findings will help improve recommendations for land managers regarding seed mixture composition and management intensity to improve effectiveness and reduce costs.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified **in one** of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;

Establishing high quality non-crop habitat patches for insect pollinators can mitigate the adverse effects of broad-scale landscape change. Our work documents that native wildflower and milkweed plantings can enhance pollinator diversity and abundance in a row crop-dominated landscape by providing food resources and nesting sites. We found that several habitat

characteristics are particularly important for increasing the ecological fitness of pollinator populations, including seed source (regional genetic differences), management through mowing, and the configuration of the habitat within the broader landscape. The US Farm Bill has invested in a range of incentive-based conservation programs on agricultural land to mitigate pollinator declines while providing multiple other benefits for agriculture and surrounding environments. Our findings will help improve recommendations for land managers regarding seed mixture composition and management intensity to improve effectiveness and reduce costs.

- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

- 2.4. What were the **goals/specific aims of the project**? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Our overall goals were to identify the important characteristics of systems that provide critical habitat for two declining pollinator groups, bumble bees and monarch butterflies; to use this knowledge to develop a model for pollinator enhancement across the state in collaboration with coordinating entities like the Missouri Department of Conservation, Missourians for Monarchs, and Missouri Master Naturalists; and to relay this information to homeowners, landowners, farmers, and agricultural consultants through web sites, social media, field days, workshops, and conferences.

To meet these goals, we had the following targets:

- 1) Completion of Objective 1: Improve the success of pollinator habitat plantings by developing recommendations on seed origin, seed mix diversity, planting density, and location of habitat to maximize the benefits for monarch butterflies and bumble bee populations.



- Study 1: *Are the flowering times of locally-collected seeds better synchronized with local pollinator populations?*

Status – field study completed; publication in preparation for submission to journal

- Study 2: *Do pollinator populations benefit from increasing the diversity and density of floral resources?*

Status – field study completed; issues with plant establishment due to drought following seeding prevented a full examination of the effects of seed density and diversity; data analysis is on-going

- Study 3: *Does the proximity of the pollinator habitat to natural habitat influence pollinator abundance? Is there a preference in the use of common milkweed vs. swamp milkweed for oviposition by monarch butterflies? Does the presence of fall nectar plants increase the use of milkweed plants for oviposition?*

Status – field study completed; manuscript submitted for publication

- 2) Completion of Objective 2: Determine proper on-going management of critical milkweed and nectar species for optimal plant establishment and growth to maximize survival and reproduction of monarch butterflies and bumble bees.

- Study 4: *Can milkweed and nectar species be managed to coincide better with monarch migration?*

Status – field study completed; data analysis on-going

- 3) Completion of Objective 3: Engage Master Gardeners and Master Naturalists across Missouri to observe and record utilization of native plant species by monarchs and bumble bees to further our understanding of the resource requirements for these species in Missouri.

- Study 5: *When do monarchs appear and reproduce in Missouri? In what types of habitats and on what types of plants are they found? Where are bumble bees found, and what plants do they visit?*

Status - established the Missouri Monarch Monitoring Team to submit monarch observations from around the state; collaborated with the Missouri Department of Conservation and the Xerces Society for Invertebrate Conservation to establish the Missouri Bumble Bee Atlas



- 4) Completion of Objective 4: Communicate knowledge gained to homeowners, landowners, and agricultural consultants to encourage community support for enhancing pollinator habitat.

Status – We participated in 39 community outreach events and produced 5 youtube informational videos

- 5) Field days

Status – We participated in 10 events held at University of Missouri Agriculture Experiment Station Farms

- 6) Participation in scientific conferences

Status – we attended 10 different scientific conferences with 8 oral paper presentations and 5 poster presentations; we also provided 4 invited seminars at various universities around the US

- 7) Data analysis

Status – data analysis for Objective 1 studies 1 & 3 is complete; data analysis for Objective 1 study 2 and Objective 2 is on-going

- 8) Manuscript preparation

Status – Study 1: in preparation; Study 2: not yet begun; Study 3: completed/submitted; Study 4: not yet begun

- 2.5. Have any of the **major goals/specific aims** or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

In 2018 the observation was made that aphids (*Aphis nerii*) are abundant on milkweeds and appear to influence monarch host plant use via changes in plant quality and attraction of natural enemies. We expanded the goal of Objective 1 Study 3 to address the potential role that aphids play in oviposition preference by adult monarchs. We added experiments to address the potential critical role that aphids play in milkweed preference by monarchs.

- 2.6. What was **accomplished under the goals/specific aims** or milestones for the



project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

OBJECTIVE 1: Improve the success of pollinator habitat plantings by developing recommendations on seed origin, seed mix diversity, planting density, and location of habitat to maximize the benefits for monarch butterflies and bumble bee populations.

Study 1 – Are the flowering times of locally-collected seeds better synchronized with local pollinator populations?

We collected seed from different ecoregions of Missouri, planted them in a common garden in central Missouri, and have tracked the flowering phenology of these perennial plants over three years. We find that seed origin (northern, central, or southern Missouri) influences flowering phenology, and this effect has carried-over for three years post-planting (time*seed origin interaction: 2020, $F_{60, 86} = 6.5$, $P < 0.001$; 2021, $F_{88, 660} = 6.74$, $P < 0.001$, 2022, $F_{32, 127} = 4.77$, $P < 0.001$). On average, peak bloom time for plants from seed collected in the Central Dissected Till Plains region (northern Missouri) was approximately 10 days prior to plants from seed collected in the Ozark Highlands or the Osage Plains; however, the exact nature of the responses differed across plant species (plant species*time*seed origin interaction: 2021, $F_{88, 352} = 7.52$, $P < 0.0001$; 2022, $F_{128, 422} = 3.81$, $P < 0.0001$). For two of the species, plants from seed collected in the *north* bloomed *earlier* than those from central or southern Missouri: swamp milkweed (*Asclepias incarnata*) and New England aster (*Symphyotrichum novae-angliae*). For one of the species, plants from seed collected in *south* bloomed *after* those from the northern or central Missouri: tall ironweed (*Vernonia altissima*). Two species showed no effect of seed origin on flowering phenology: common milkweed (*Asclepias syriaca*) and prairie blazing star (*Liatris pycnostachya*). We hypothesize that differences in environmental cues (e.g., temperature, sunlight, etc.) that accompany a shift to the south for the northern seed or a shift to the north for the southern seed are enough to impact the flowering phenology of plants from different ecoregions. In this case, seeds were moved only ~200 miles from the origin, but this distance is enough to result in an ~2° F difference in average daytime and nighttime temperatures. We expect that not all species responded similarly due to genetic differences in cues that are important for phenology.

We have tracked bumble bee visitation to plants grown from seed collected from different ecoregions over the past few years. We see consistent responses of



bumble bees across years; however, the abundance of bees was very low in 2022 due to extreme drought conditions. As a result, we share our results from 2022 with caution. Despite this, we find that the frequency of bumble bee visitation and the composition of the community did not differ in response to seed origin in 2021 or 2022 ($F_{2, 12} = 0.56, P > 0.58$ and $F_{2, 12} = 1.09, P > 0.37$). In 2021 and 2022, flowering phenology influenced the timing of bumble bee visitation to seeds from different regions (week * seed origin interaction: $F_{12, 72} = 4.08, P < 0.0001$; $F_{24, 137} = 2.41, P < 0.001$).

We propose that for generalist pollinators like bumble bees a seed mix including plants from different regions with different flowering times may prolong the bloom period and provide extended resources to pollinators. We suspect that the response may not be the same for more specialized bees with more specific flowering needs. In this case, shifts in flowering phenology may result in asynchrony with local specialist bees.

Study 2 – Do pollinator populations benefit from increasing the diversity and density of floral resources?

We manipulated the density and diversity of floral resources in a conservation seed mix using a factorial design and monitored plant establishment, bloom phenology, and bumble bee visitation in subsequent years. This experiment is the only study we conducted in which native plant plots were established from seed directly applied in the field (plugs from plants grown in the greenhouse were used for all other studies). Establishing plots from seed that reflect our intended treatments has proven to be a challenge. The major impact of increasing the density and diversity of seeds in the mix on pollinators (specifically bumble bees) is to increase the likelihood that partridge pea (*Chamaecrista fasciculata*) is present in a plot. This "identity effect" of diversity is due to the higher establishment rate of partridge pea than other native species and the attractiveness of partridge pea to bumble bees. We observed no differences in the pollinator communities associated with treatment, but the vast majority of bumble bees were observed foraging on the dominant partridge pea. Six species of bumble bees were observed foraging in the plots, including the Southern Plains Bumble Bee (*Bombus fraternus*), which has been declining in abundance. At this point in the study, we see no economic benefit to increasing seed planting density beyond 25 pure live seeds/ft².

Study 3 – Does the proximity of pollinator habitat to natural habitat influence monarch butterfly abundance? Is there a preference in the use of common milkweed vs swamp milkweed for oviposition? Does the presence of fall nectar plants increase the use of milkweed plants for oviposition?

Monarch abundance in our sampling location was extremely low for summer 2022 & 2023 (almost non-existent). The number of monarch larvae and eggs observed were so low that we are unable to perform formal statistical analyses on the data.



We used structural equation modeling (SEM) with data collected in 2019-2021 to examine how environmental factors, including landscape context and the presence of predator and competitor insects, affect monarch egg abundance on milkweed plants. Some takeaways from the SEM:

- *No evidence that wildflowers reduce the apparency of milkweed hosts to monarchs.* The presence of nectar plants surrounding milkweed plots did not affect monarch egg abundance on milkweeds. As long as nectar plants were present in the general area, it did not matter whether milkweeds were planted in a wildflower matrix or adjacent to a wildflower plot. Monarchs were able to locate milkweed host plants, even among a heterogeneous mixture of plants.
- *There was a positive effect of the presence of trees on monarch egg abundance.* Although tree lines compete with native plants for water, wooded areas may help increase monarch egg abundance by alleviating the microclimate and providing adults with a refuge.
- *There is a positive association between oleander aphids and monarch eggs.* Others have shown that oleander aphids induce defensive chemistry changes in the host plant that benefit larval monarchs by short-circuiting plant defense against larvae. We hypothesize that adult monarchs may detect these changes in plant quality and make their oviposition decisions accordingly. Monarch eggs may also gain *associational resistance* from natural enemies when alternative prey (aphids) are present.
- *Predatory lady beetles track their aphid prey.* Lady beetles exhibited a numerical response to the presence of oleander aphids on milkweed plants.

OBJECTIVE 2: Determine proper on-going management of critical milkweed and nectar species for optimal plant establishment and growth to maximize survival and reproduction of monarch butterflies and bumble bees.

Study 4 – Can milkweed and nectar species be managed to coincide better with monarch migration?

We investigated the effects of mowing on utilization of plantings by monarchs. We managed plots of milkweed and nectar plants with three treatments: control with no management, mow to 15 cm at the end of May, or mow to 15 cm at the end of June. We observed that mowing resulted in smaller but more tender milkweed tissue available for monarch larvae later in the season. There was potentially slightly higher monarch egg deposition on milkweed plants that were mowed than those that were not. Flowering time of nectar plants appeared to be delayed by mowing, but there were no clear differences in monarch or bumble bee visitation. In 2022, severe drought conditions limited plant re-growth following mowing and resulted in fewer milkweed resources available to monarchs at the end of the season. However, monarch abundance in 2022 was extremely low throughout the year, so it is unclear what effect this habitat manipulation had on the population.



OBJECTIVE 3: Engage Master Gardeners and Master Naturalists across MO to observe and record utilization of native plant species by monarchs and bumble bees to further our understanding of the resource requirements for these species in MO.

We initiated the Missouri Bumble Bee Atlas community science program in collaboration with the Xerces Society, the Missouri Department of Conservation, and Pheasants/Quail Forever. Volunteers received online and in person training in non-lethal methods to capture and document bumble bee habitat use during 45-minute surveys. Volunteers are assigned one of 75 grid cells within the state, ensuring statewide coverage. Data is uploaded to the ATLAS website and analyzed to determine species and flower/habitat preferences. Eight species of bumble bees were found, including three Species of Greatest Conservation Need (SGCN) and one Species of Conservation Concern (SOCC). Half-black bumble bees (*Bombus vagans*) had not been documented within the state since 2015 and is not recorded in iNaturalist for Missouri; seven individuals including several new county records were documented. Yellow bumble bee (*Bombus fervidus*) had not been recorded in Missouri in the past, but one individual was documented in Buchanan County.

We initiated the Missouri Monarch Monitoring Team, which was active for two years. Community scientists provided monarch observations every two weeks from May through September. Observation locations were concentrated along the I-70 corridor in central Missouri, but also included sites in southern and northern Missouri. A survey of participants found that:

- 93% found the experience rewarding.
- 87% agreed or strongly agreed that participation in this project motivated them to learn more about pollinators.
- 87% agreed or strongly agreed that participation in this project motivated them to plant more pollinator habitat.
- 67% made an observation of the natural world that they had not noticed before

Objective 4: Communicate knowledge gained to homeowners, landowners, and agricultural consultants to encourage community support for enhancing pollinator habitat.

We provided 49 presentations to the public and produced 5 YouTube videos. (See details in section 2.9 below.)

KEY OUTCOMES:

Our data suggest that pollinator habitats can be planted and managed in ways that enhance the utility of habitats for monarchs and bumble bees. Use of seed from adjacent regions and management through mowing may extend the time that resources are available to pollinators without disrupting synchrony. Milkweed planted

near shady, wooded areas may benefit adult monarchs without increasing the risk of predation on monarch eggs. Milkweed plants within a matrix of diverse flowering nectar plants are equally apparent to monarch females as those in stand-alone plots.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

Methods were developed following a search of the scientific literature and consultation with experts at the Missouri Department of Conservation. Feedback from the broader research community was solicited during presentations of preliminary work at scientific meetings. Colleagues serving on graduate committees provided input on design, implementation, and analysis of experiments. A statistician and a data scientist were consulted on some of the analyses.

- 2.8. List key stakeholders that could be served by results of this project.

Farmers participating in incentive-based Farm Bill conservation plantings
Conservation land managers who work with farmers
Community members interested in promoting pollinators

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Outreach presentations:

Date	Location	Event name	Event format
2018			
17-Mar-18	Columbia, MO	2018 Mid-Missouri Home Show	Class for registered participants
20-Mar-18	Boone Co., MO	Boone Co. Master Gardeners/Naturalists	~1.5 hours talk, Boone Co. extension office
31-Mar-18	Rolla, MO	Master Naturalists/ Master Gardeners	talk
20-Jul-18	Columbia, MO	High school students from Brazil	talks
15-Sep-18	Weldon Springs, MO	Monarch Madness	talk/booth
18-Sep-18	Boone Co., MO	FFA students at Bradford Farm	talks
29-Sep-18	Columbia, MO	South Farm Showcase	booth/hands on
14-Nov-18	Columbia, MO	Grow Native! Conference	poster
2019			
18-Feb-19	Springfield, MO	SW Missouri Naturalists	Chapter meeting
13-Apr-19	Columbia, MO	Bradford Farm Native Plant Sale	booth
13-Apr-19	Columbia, MO	Show Me Mizzou Day	booth
27-Apr-19	Columbia, MO	2019 Mid-Missouri Home Show	Class for registered participants
19-Apr	Columbia, MO	The Conservation Agent Talk Show on KOPN	radio interview
28-Apr-19	Columbia, MO	Earth Day Celebration	booth

19-Jun-19	Columbia, MO	Native pollinator habitat workshop	Training for MO Dept of Conservation
20-Jun-19	Columbia, MO	Missouri Pollinator Week	Tours of research plots
10-Jul-19	Kingdom City, MO	Youth Monarch Conservation Training	In conjunction with the juvenile justice system
30-Jul-19	Columbia, MO	MU Turf field day	presentation to turf and landscape managers
19-Jul-19	Columbia, MO	Mizzou Academy	talk to students from Brazil, South Korea, US
7-Aug-19	Columbia, MO	Youth Monarch Conservation Training	In conjunction with the juvenile justice system
14-Aug-19	Holt's Summit, MO	Master Gardeners	chapter meeting
18-Sep-18	Boone Co., MO	FFA high school students	talks
28-Sep-19	Columbia, MO	South Farm Showcase community event	booth/hands on

2020

5-Mar-20	Columbia, MO	Community screening of film <i>Hometown Habitat</i>	member of expert panel for discussion following screening
20-Oct-20	Facebook - MU Central Missouri Research, Extension and Education Center	University of Missouri South Farm Virtual Field Day	https://www.facebook.com/watch/?v=1252553555116661

2021

24-Jun-21	Columbia, MO	Pollinator event, Jefferson Farm & Garden	booth and tour of research plots
25-Jun-21	Columbia, MO	Missouri Master Pollinator Steward	Training for short course



20-Jul-21	Columbia, MO	Heart of Missouri Master Gardeners	pollinator research presentation
4-Aug-21	Novelty, MO	FFA high school students	talks
24-Sep-21	Columbia, MO	Missouri Master Naturalists	training for beginner pollinator course

2022

4-Jun-22	Ballwin, MO	Community pollinator event	booth
23-Jun-22	Columbia, MO	Pollinator event, MU South Farm	booth and tour of research plots
17-Jul-22	Columbia, MO	Boone Co. Honeybee Club	talk presentation on pollinators
21-Jul-22	Columbia, MO	Girl Scout Troop	talk to students from Brazil, US
22-Jul-22	Columbia, MO	Mizzou Academy FFA high school students	talks
4-Aug-22	Boone Co., MO	South Farm Home & Garden Showcase	booth
27-Aug-22	Columbia, MO	Locust Street Expressive Arts Elementary	pollinator lesson lecture to biology students
4-Oct-22	Columbia, MO		
12-Oct-22	Point Lookout, MO	College of the Ozarks	

2023

12-Feb-23	Boone Co., MO	Boone Co. Extension Office	pollinator presentation
15-Feb-23	Columbia, MO	City of Columbia Stormwater Management	pollinator discussion expert panel member
4-May-23	Columbia, MO	Missouri Master Pollinator Stewardship Program	training presentation



11-May-23	Columbia, MO	Hallsville Elementary	pollinator presentation
17-Jun-23	Columbia, MO	Butterfly Festival @ Jefferson Farm	prairie walk and booth
26-Jun-23	online	Missouri Bumble Bee Atlas	online training
8-Jul-23	Jefferson City, MO	Missouri Bumble Bee Atlas	in person training
21-Jul-23	Columbia, MO	Mizzou Academy	talk to students from Brazil, US
30-Sep-23	Columbia, MO	CAFNR Showcase at South Farm	insect booth
10-Dec-23	online	Missouri Bumble Bee Atlas	online training

Online videos:

Five pollinator videos were posted to the Missouri Ag Experiment Station YouTube channel: <https://www.youtube.com/@MOAES/videos>

- Establishing pollinator habitat
- Superior pollinator habitat
- Pollination habitat rehabilitation and bees
- Monarch butterfly decline and what we can do

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

There were three main challenges that were resolved through the awarding of a one-year no cost extension:

- 1) Covid-19: The University of Missouri instituted a hiring freeze in 2020 due to Covid-19. The freeze resulted in carry-over issues with having enough personnel to manage the proposed work.

- 2) 2022 Drought: Boone County, Missouri experienced severe drought in 2022, and we were under a state-issued 'drought alert' for extreme heat and dry conditions since July. The extreme environmental conditions resulted in poor plant growth and limited flower production in our experimental plots in 2022. The lack of flowers precluded an assessment of pollinator responses to treatments that year.
 - 3) 2022 Low Monarch Numbers: Despite a 35% increase in the overwintering population of monarchs in Dec. 2021, monarch abundance in our sampling location was extremely low for summer 2022 (almost non-existent). The number of monarch larvae and eggs observed this summer were so low that we are unable to perform formal statistical analyses on the data.
- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Two graduate students, 1 research specialist, and 10 undergraduates were supported on this project during the reporting period. The goal of the mentoring activities is to provide the skills, knowledge, and experiences necessary for the students to become independent researchers who are prepared to excel in their chosen career path. Students identify career and professional goals that are used to develop tailored experiences. The students and PI engage in regular conversations about research projects, professional advancements, and the diverse career opportunities available in the field. The students receive guidance on finding career opportunities, networking, applying for positions, establishing a research program, and maintaining work-life balance. Graduate students and the PI meet for a formal evaluation, feedback, and assessment each year. This is a two-way conversation that provides an opportunity for the student to also provide feedback to the mentor. All of the students are encouraged to present the results of their work at scientific meetings. One of the undergraduate students received first place in the Undergraduate Student Poster competition and both graduate students received second place in the Graduate Student Oral Presentation competition. Undergraduate and graduate students are also actively engaged in the planning and execution of all outreach events and online videos.



- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

2 graduate students, 10 undergraduate students

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Scholarly publications, scientific presentations

- 3.2. Please provide a list of citations for the information products produced during the project.

Scholarly publications:

Harris, K.L., D. Hall, and D.L. Finke. 2023. Who cares about monarch butterflies? Comparing US State Wildlife Action Plans 2015–2025. *Conservation Letters* 16: e12976.

Harris, Katie, Tim Reinbott, Terry Woods, Jared Brabant, Grant Harris, and Deborah Finke. 2024. Context-dependency of monarch butterfly (*Danaus Plexippus*) egg abundance on milkweeds (*Asclepias*). Submitted for publication.

Brabant, Jared, Tim Reinbott, Terry Woods, Katie Harris, and Deborah Finke. Influence of seed ecotype on wildflower plot establishment, phenology, and pollinator visitation. In preparation for publication.

Scientific presentations:

Woods, T, T. Reinbott, K. LaPlante, and D. Finke. 2019. Bumble bee responses to native wildflower plantings in Missouri. Poster presentation. Entomological Society



of America National Meeting, St. Louis, MO.

LaPlante, K., T. Woods, T. Reinbott, and D. Finke. 2019. Influence of habitat on monarch butterflies (*Danaus plexippus*), their competitors, and natural enemies. Poster presentation. ESA-North Central Branch meeting.

LaPlante, K., T. Woods, T. Reinbott, and D. Finke. 2019. Monarch butterfly (*Danaus plexippus*) host plant selection and the effects of aphid competitors. Oral presentation. Entomological Society of America National Meeting, St. Louis, MO

Monarch butterfly (*Danaus plexippus*) host plant selection and the effects of aphid competitors. K. LaPlante, T. Woods, T. Reinbott, and D. Finke. 2020. ESA-North Central Branch virtual meeting.

Landscape and community context influence monarch butterfly (*Danaus plexippus*) egg abundance. K. LaPlante, T. Woods, T. Reinbott, and D. Finke. 2019. Entomology Society of America national virtual meeting.

LaPlante-Harris, Katie, Terry L. Woods, Timothy M. Reinbott and Deborah L. Finke. Oleander aphid dynamics affect monarch butterfly egg abundance on milkweed host plants. Poster presentation at Ecological Society of America Annual Meeting. August 2021.

Brabant, Jared, Katie LaPlante-Harris, Terry L. Woods, Timothy M. Reinbott and Deborah L. Finke. Expansion of the conservation toolkit: using seed provenance diversity to support durability of bumble bee (*Bombus* spp.) resources. Oral presentation at Entomological Society of America, November 2021.

Katie LaPlante-Harris, Brabant, Jared, Terry L. Woods, Timothy M. Reinbott and Deborah L. Finke. Oleander aphids (*Aphis nerii*) mediate a shift in monarch butterfly (*Danaus plexippus*) host plant selection. Oral presentation at Entomological Society of America, November 2021.

Finke, Deborah. Insect ecosystem services in a changing world: pollinator responses to habitat reconstruction. Seminar to the Department of Entomology at Purdue University. November 17, 2021.

Brabant, Jared and Debbie Finke. The contribution of intraspecific floral diversity to season-long availability of bumble bee resources. Oral presentation at Entomological Society of America, Vancouver, November 2022.

Critchfield, Emma, Katie Harris, Debbie Finke. Milkweed species identity mediates the effects of drought and parasitism on oleander aphids (*Aphis nerii*). Poster presentation at Entomological Society of America, Vancouver, November 2022.

Harris, Katie, Damon Hall and Debbie Finke. Who cares about monarch butterflies? Comparing US State Wildlife Action Plans 2015-2025. Oral presentation at Entomological Society of America, Vancouver, November 2022.

Finke, Debbie. Insect ecosystem services in a changing world. Invited oral presentation, CAFNR Research Day. October 12, 2022.

Finke, Debbie. Insect ecosystem services in a changing world. Seminar to the



Department of Entomology at the University of Florida. December 2, 2022.

Finke, Debbie. Insect ecosystem services in a changing world. Seminar to the Department of Entomology at Kansas State University. December 6, 2022.

Brabant, Jared and Debbie Finke. 2023. Bumble bee community response to Farm Bill plantings in Missouri. Poster presentation. Entomological Society of America Annual Meeting.

Woods, Terry, Tim Reinbott, and Debbie Finke. 2023. Missouri milkweed (*Asclepias*) distribution change over ~40 years. Poster presentation. Agronomy Society of America Annual meeting.

Harris, Katie, Damon Hall, and Debbie Finke. Who cares about monarch butterflies? Comparing US State Wildlife Action Plans. Poster presentation. American Association for the Advancement of Science (AAAS) meeting.

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Harris, K.L., D. Hall, and D.L. Finke. 2023. Who cares about monarch butterflies? Comparing US State Wildlife Action Plans 2015–2025. *Conservation Letters* 16: e12976.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

None

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	



Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

None

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

Published (or will be published) in peer reviewed literature, uploaded to university website

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Visual observations of pollinator visitation
Phenological data for native plants
Photographs of bumble bees and native plant plots
Site location and description data and monarch observations from citizen scientists

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide

a short description of how they are related.

Data sets are independent

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Title [name of dataset]	Treatment design	Description [short narrative description]
Density_Diversity Data	Factorial manipulation of wildflower density and diversity	Observations of blooming plant species and bee visitations to flowers
Ecotype Data	Wildflower seed collected in north, central, and south MO and planted in a common garden	Plant establishment, bloom phenology, and bumble visitation
Management Data	Manipulated field management: control, mow to 15 cm at the end of May, or mow to 15 cm at the end of June.	Plant bloom phenology, bumble bee visitation, monarch egg abundance, aphid abundance
Monarch mail data	Community science monarch monitoring	Participant list, habitat descriptions, monarch observations over time
Monarch plots	Proximity to natural habitat, milkweed species identity, nectar plant presence	Monarch egg abundance, aphid abundance, natural enemy abundance, milkweed bloom phenology, milkweed height



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Plot pictures	Proximity to natural habitat, milkweed species identity, nectar plant presence	Photographs of plots over time
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Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: _____ Deborah L. Finke _____

PI Signature: _____ *Deborah L. Finke* _____ Date: _____ 3/26/2024 _____

Authorized Signing Official (ASO) Name: _____ Chad Hampton _____

ASO Signature: _____ *Chad Hampton* _____ Date: _____ 04/01/2024 _____

Digitally signed by Chad Hampton
Date: 2024.04.01 16:06:25 -05'00'